

Code-along 04

FirstName LastName

Setup

Packages

Install the `Rmisc()` package, available on CRAN.

```
install.packages("Rmisc")
```

Load the standard packages and our new `Rmisc()` package.

```
library(tidyverse)
library(haven) # not core tidyverse
library(gssr)
library(gssrdoc) # load the GSS codebook
library(summarytools)
library(Rmisc) # load the new package
```

Load your data

Get the data from the 2022 survey

```
# ADD YOUR CODE HERE
```

Create a dataset named `my_data` with the variables `wwwhr`, `postlife`, `childs`, & `sex` with no missing data

```
# ADD YOUR CODE HERE
```

Confidence Intervals

`summarise()` with equation

```
wwwhr_sum <- my_data |>
  dplyr::summarise(
    n = n(),
    mean_wwwhr = mean(wwwhr),
    sd_wwwhr = sd(wwwhr),
```

```
se_wwhr = sd_wwhr / sqrt(n),
ci_lower = mean_wwhr - (1.96 * se_wwhr),
ci_upper = mean_wwhr + (1.96 * se_wwhr)
)
```

```
Error:
! object 'my_data' not found
```

Code Meets Question

What can we infer about the population mean based on this confidence interval?

```
wwhr_sum
```

```
Error:
! object 'wwhr_sum' not found
```

summarise() + CI()

```
wwhr_ci <- my_data |>
  dplyr::summarise(
    n = n(),
    mean_wwhr = mean(wwhr),
    sd_wwhr = sd(wwhr),
    lowCI_wwhr = Rmisc::CI(wwhr)['lower'],
    hiCI_wwhr = Rmisc::CI(wwhr)['upper']
  )
```

```
Error:
! object 'my_data' not found
```

```
wwhr_ci
```

```
Error:
! object 'wwhr_ci' not found
```

summarise() + CI()

```
wwhr_ci99 <- my_data |>
  dplyr::summarise(
    n = n(),
```

```

mean_wwhr = mean(wwhr),
sd_wwhr = sd(wwhr),
lowCI_wwhr = CI(wwhr, ci = 0.99)['lower'],
hiCI_wwhr = CI(wwhr, ci = 0.99)['upper']
)

```

Error:
! object 'my_data' not found

```
wwhr_ci99
```

Error:
! object 'wwhr_ci99' not found

How does a 95% confidence interval differ from a 99% confidence interval in terms of width and certainty?

CI (wrong) with proportions

```

my_data |>
  dplyr::summarise(
    n = n(),
    mean_ghosts = mean(postlife),
    sd_ghosts = sd(postlife),
    lowCI_ghosts = CI(postlife, ci = 0.99)['lower'],
    hiCI_ghosts = CI(postlife, ci = 0.99)['upper']
  )

```

Error:
! object 'my_data' not found

CI (correct) with proportions

```

# Use logic to recode (yes: 1 = 1; no: 2 = 0)
my_data$ghosts <- ifelse(my_data$postlife == 1, 1, 0)

```

Error:
! object 'my_data' not found

```
my_data |>
  dplyr::summarise(
    n = n(),
    mean_ghosts = mean(ghosts),
    sd_ghosts = sd(ghosts),
    lowCI_ghosts = CI(ghosts, ci = 0.99)['lower'],
    hiCI_ghosts = CI(ghosts, ci = 0.99)['upper']
  )
```

Error:
! object 'my_data' not found

Comparing CIs

```
my_data$sex <- zap_missing(my_data$sex)
```

Error:
! object 'my_data' not found

```
my_data$sex <- as_factor(my_data$sex)
```

Error:
! object 'my_data' not found

```
my_data$sex <- droplevels(my_data$sex)
```

Error:
! object 'my_data' not found

```
df_ghosts <- my_data |>
  group_by(sex) |>
  dplyr::summarise(
    n = n(),
    mean = mean(ghosts),
    sd = sd(ghosts),
    lower = CI(ghosts, ci = 0.95)['lower'],
    upper = CI(ghosts, ci = 0.95)['upper']
  )
```

```
Error:
! object 'my_data' not found
```

```
df_ghosts
```

```
Error:
! object 'df_ghosts' not found
```

```
df_wwhr <- my_data |>
  group_by(sex) |>
  dplyr::summarise(
    n = n(),
    mean = mean(wwhr),
    sd = sd(wwhr),
    lower = CI(wwhr, ci = 0.95)['lower'],
    upper = CI(wwhr, ci = 0.95)['upper']
  )
```

```
Error:
! object 'my_data' not found
```

```
df_wwhr
```

```
Error:
! object 'df_wwhr' not found
```

Hypothesis Testing

One sample **t.test()** for means

```
t.test(my_data$childs)
```

```
Error:
! object 'my_data' not found
```

H0 = 0 by default

One sample `t.test()` for means

```
t.test(my_data$childs, mu=2, alternative="less") # Ho: mu=2
```

```
Error:  
! object 'my_data' not found
```

H0 = 2; one-tail test

two sample `t.test()` in action

Are the proportions of men and women who believe in ghosts statistically significantly different?

```
t.test(my_data$ghosts ~ my_data$sex, alternative = "two.sided")
```

```
Error:  
! object 'my_data' not found
```

Assuming an α level of .05, what do you conclude?

Aside: paired `t.test()`

Independent group t-test

Do two subsamples have the same mean?

```
t.test(my_data$ghosts ~ my_data$sex, alternative = "two.sided")
```

```
Error:  
! object 'my_data' not found
```

Paired t-test

Do two variables have the same mean?

```
t.test(Pair(hrs1_2018, hrs1_2020) ~ 1, data = gss_panel)
```

Think Like a Statistician

(Use the 2022 GSS dataset.)

Do women have more years of education (`educ`) than men, on average?

```
# YOUR CODE HERE
```

Are the proportions of men & women who support abortion for any reason ([abany](#)) statistically different?

```
# YOUR CODE HERE
```

Do fewer men than women support gun laws ([gunlaw](#))?

```
# YOUR CODE HERE
```